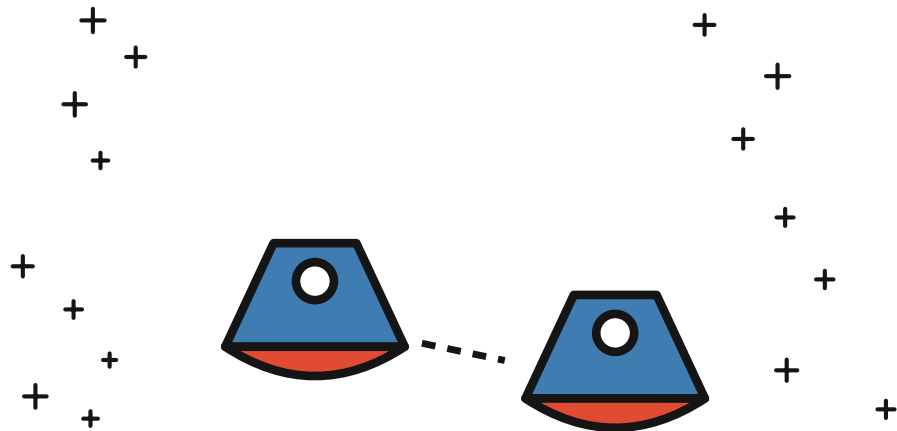


Next:

Finding Each Other

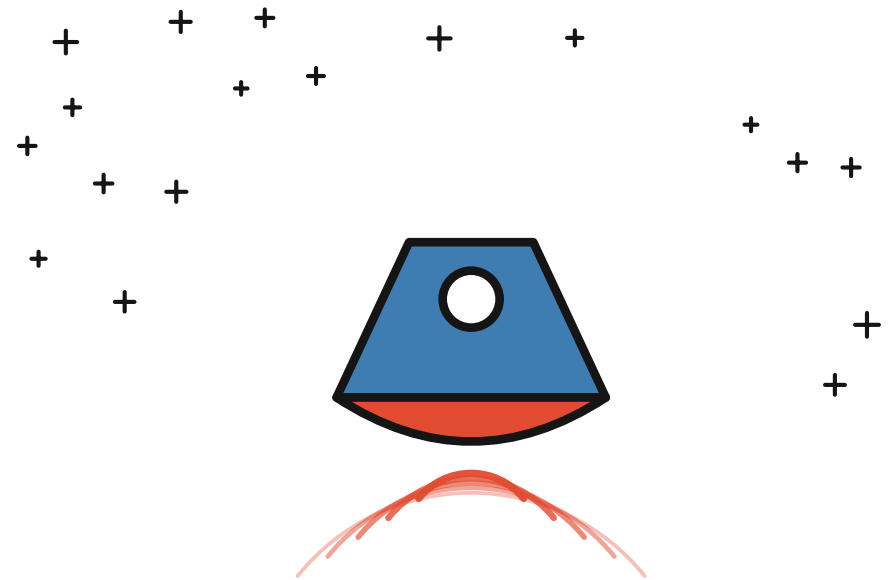
Book Seven · how two ships meet when both are falling



To catch up with something,
you have to slow down.

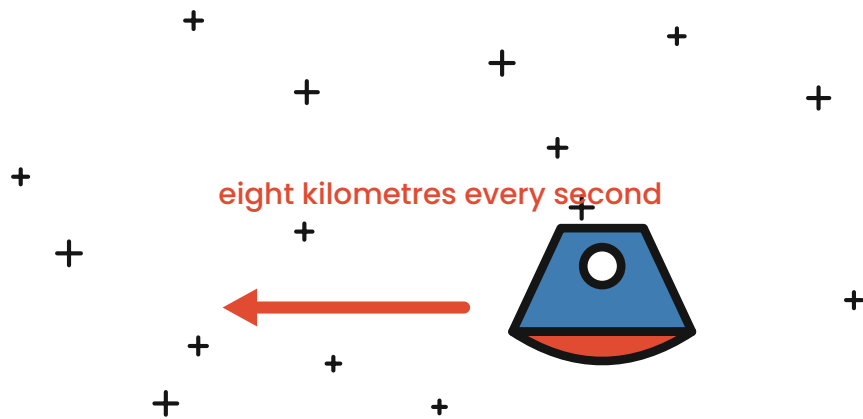
COMING HOME

Book Six · Why a blunt shape is safer than a sharp one



Print it. Read it together. Then rub your hands.

Up there, you are going very fast.



To get home, all of that has to go away.



For the grown-up.

The one idea

Re-entry is a braking problem, not a heat problem — the heat is a symptom of shedding orbital energy. A returning capsule carries roughly the same energy it took to put it up there, and essentially all of it is dumped into the atmosphere in a few minutes.

Why blunt beats pointy

H. Julian Allen at NACA showed in 1951 that a blunt body pushes a detached bow shock ahead of itself, so most of the energy heats the air rather than the vehicle. A sharp nose keeps the shock attached and delivers the heat straight to the structure. This is why every crewed capsule since has come home base-first.

Ablation

The shield is designed to char and erode. Material carrying away heat as it leaves is far more effective than trying to insulate. Apollo's shield lost several centimetres of thickness on the way home, exactly as intended.

Gagarin

12 April 1961, Vostok 1: one orbit, 108 minutes. The capsule had no landing rockets, so he ejected at about 7 km and came down under his own parachute — a detail the Soviet Union concealed for years because flight records required a pilot to land inside the craft.

Try this tonight.

1 Rub your hands together.

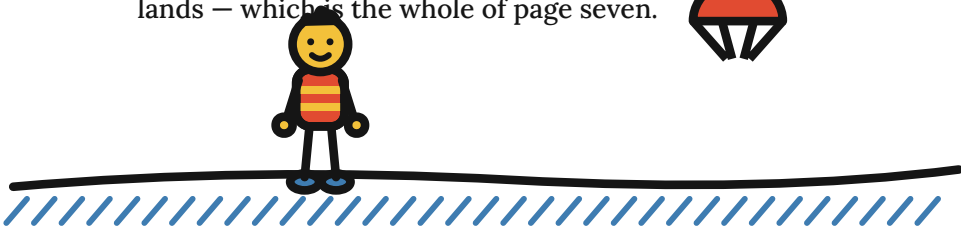
Fast, for ten seconds. They get warm, and nothing is burning — the heat comes from rubbing alone. A spacecraft rubs against the whole atmosphere at eight kilometres a second.

2 Drop a flat card and a pencil.

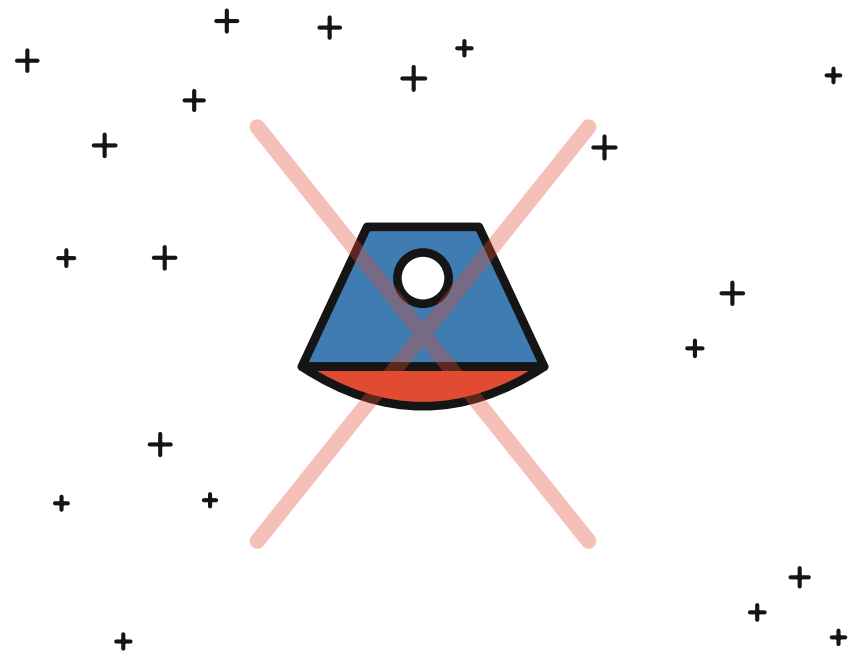
The card flutters slowly; the pencil drops straight. Broad and blunt catches the air, narrow and sharp slips through it. Coming home, you want to be the card.

3 Blow across your hand, then at it.

Blowing straight at your palm feels sharper than blowing across it. Shape decides where the push lands — which is the whole of page seven.



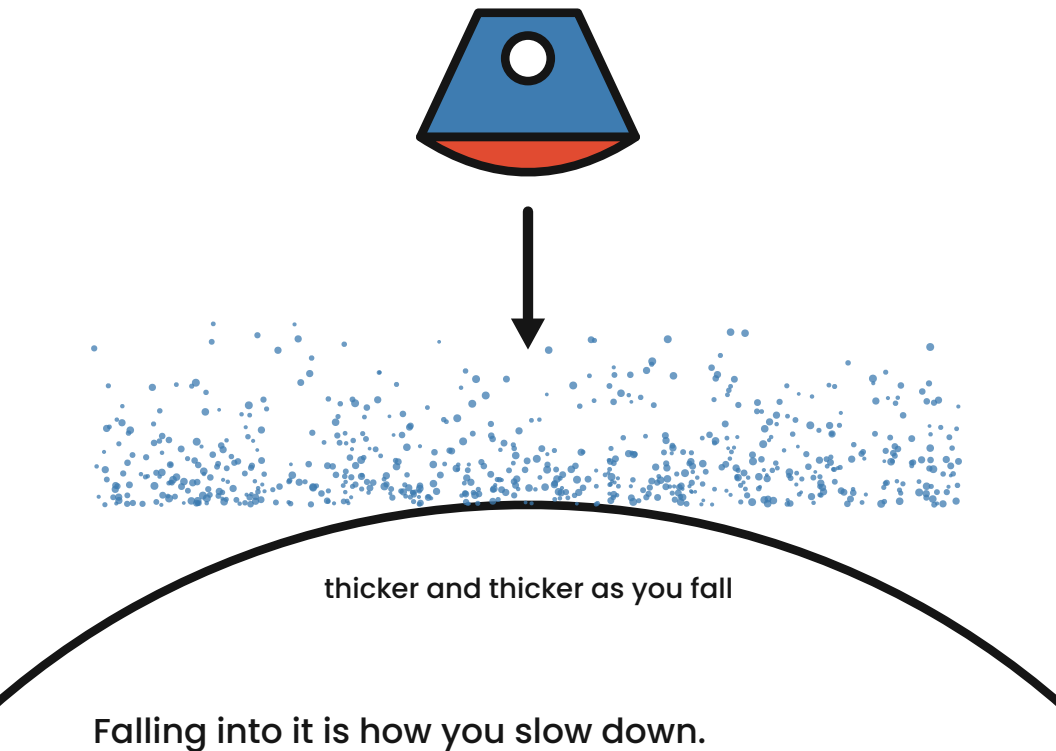
But there are no brakes in space.



Nothing to push on. Nothing to grab.
So what can slow you down?

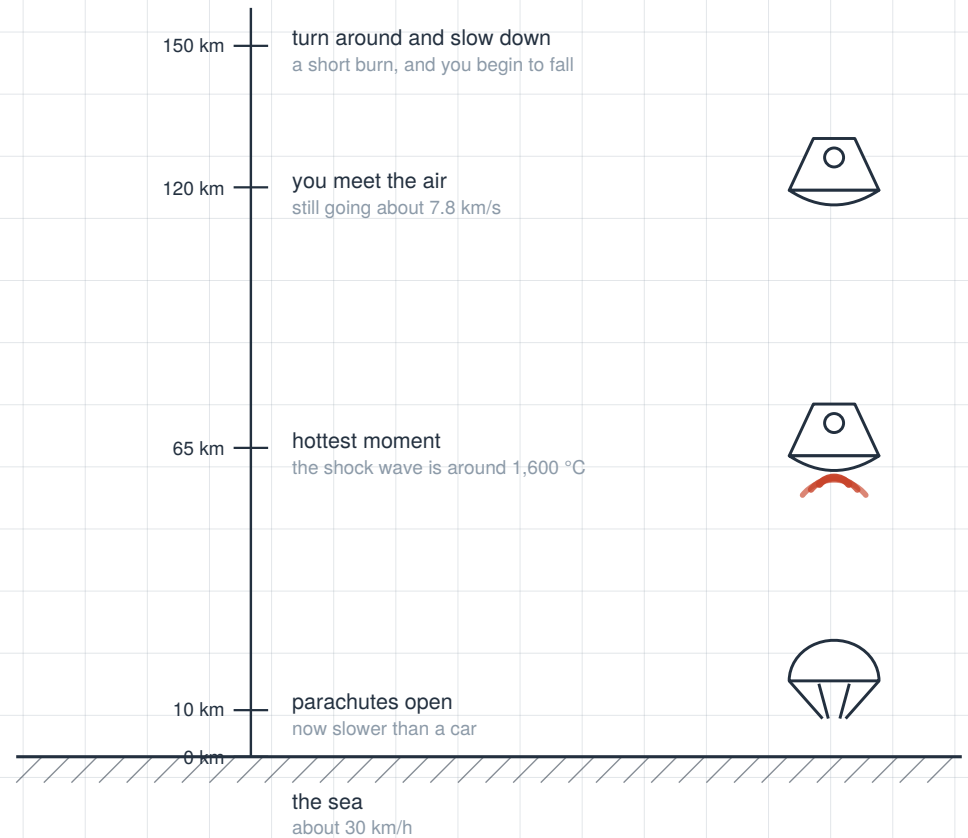
The air can.

The same air that ran out on the way up.



The way down.

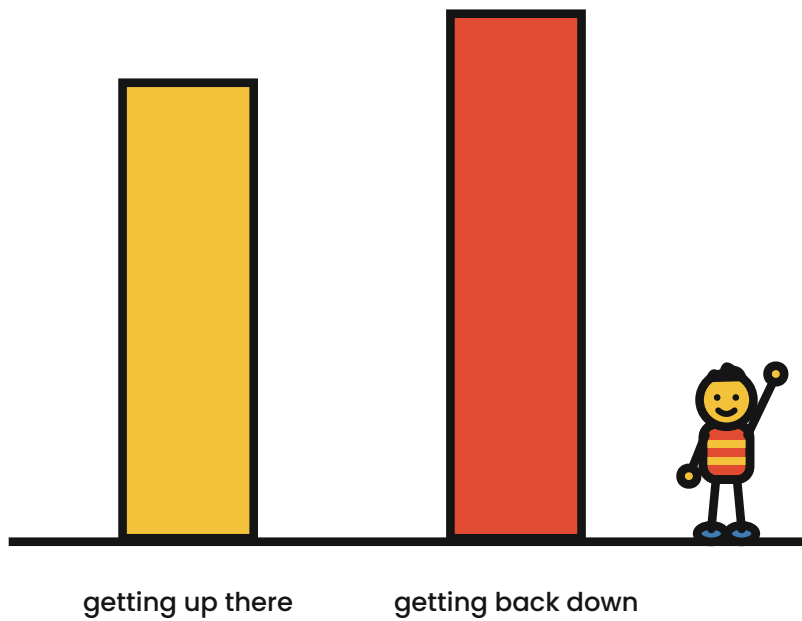
Coming Home · figure 2 · from orbit to the sea



Almost all the slowing down is done by air, not by engines.
The heat shield is the brake, and it is consumed doing the job.

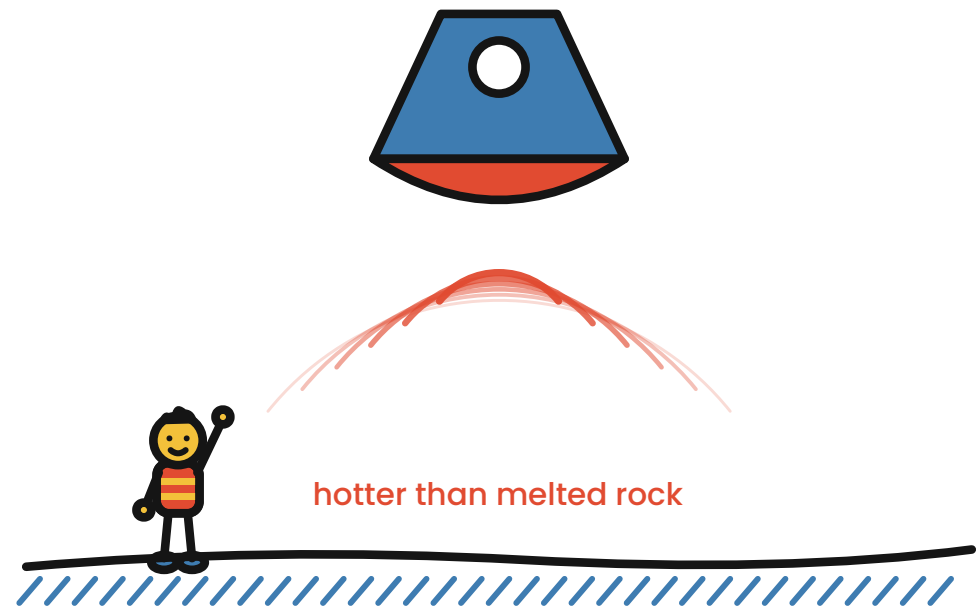
Apollo came back from the Moon faster still — about 11 km/s.

Going up is hard.
Stopping is harder.



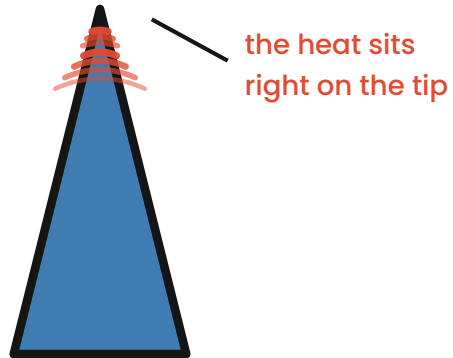
All the speed you gained, you have to give back.

But at that speed,
air is not soft.



Rub your hands together fast. Now imagine that, times a million.

**A sharp point
cuts straight in.**



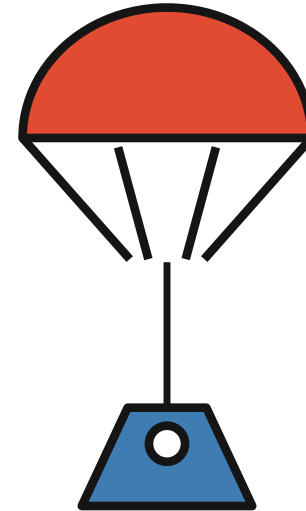
the heat sits
right on the tip

and it burns

Sharp seems right. It is exactly wrong.

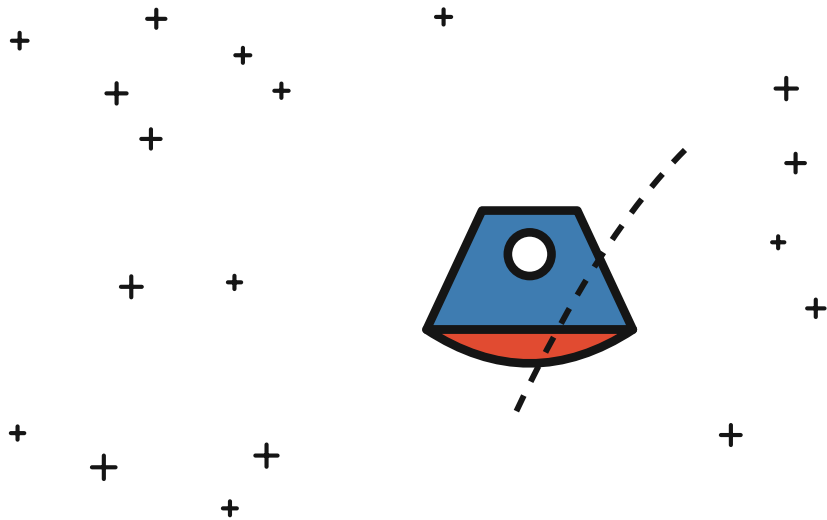
Then a parachute.

The slow, quiet part.



And then the sea, and a boat, and home.

The first person home was Yuri Gagarin.

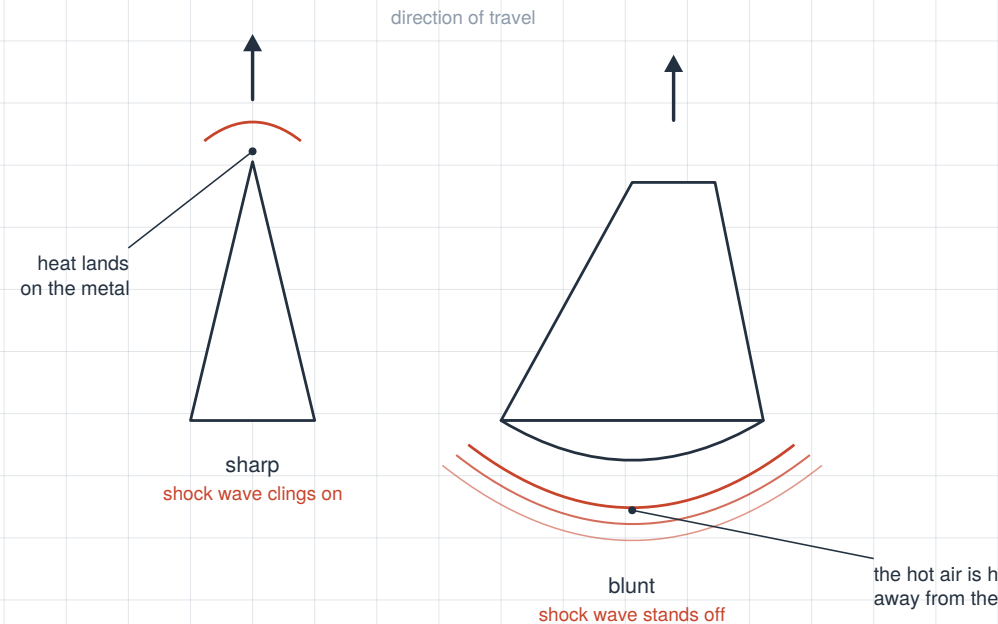


He went round the Earth one time,
and was back in under two hours.



Sharp, or blunt?

Coming Home · figure 1 · where the heat goes

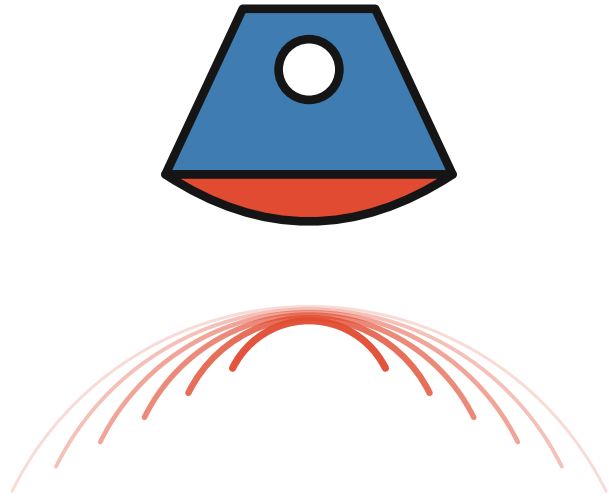


A blunt body pushes a shock wave out in front of it. Most of the energy goes into heating that air rather than the spacecraft.

H. Julian Allen worked this out in 1951. Every capsule since has come home bottom-first because of it.

**So make it fat
and round instead.**

the hot air is pushed out in front



The fire never touches you.

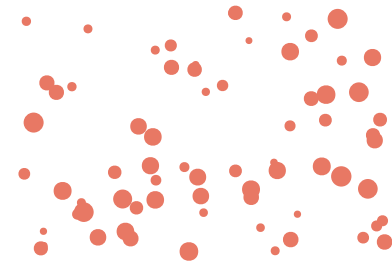
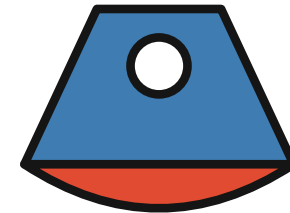
8

**And the bottom is
meant to burn away.**

pierce here

pierce here

pierce here



little bits flake off, and take the heat with them

It is not broken. It is doing its job.

9

fold along this line